

Predicting *STEM Undergraduates'* Intentions to *Pursue Scientific Research Careers*

STATS 202 Final Presentation

2025/12/01

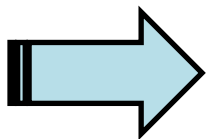
Shuhan (Alice) Ai

Introduction

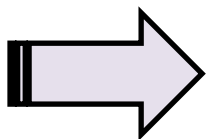
Background

- Call for expanding the number of sciences, technology, engineering, and mathematics (STEM) degrees produced.
- Faculty mentorship, undergraduate research (UGR) programs, that can strengthen students' interest, skills.
- Linear regression approaches vs. non-linear relationships, nonparametric methods.

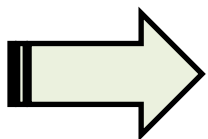
Research Questions



RQ1 (Descriptive): How do scientific career aspirations, undergraduate research time, and time allocation patterns vary across different STEM majors?



RQ2 (Regression): What relationships exist between predictors and scientific research career aspirations? Do students' undergraduate research experiences correlate with science-related psychological traits?



RQ3 (Prediction): How accurately can machine learning algorithms (logistic regression, decision trees, random forests, K-nearest neighbors, and XGBoost) predict students' likelihood of pursuing science research careers, and which features are most important across different models?

Data

• Dataset

- Student-level data: Higher Education Research Institute's ([HERI](#)) **2019 College Senior Survey** (CSS)
- Administered at the last quarter/semester when undergraduate seniors

• Study Sample

- Students who graduate with a STEM bachelor's degree
- 3,346 STEM undergraduate students
- Average missingness rate was less than 1.6%
- Removed 22 outliers $|z| > 3.0$

• Variables

– Outcome variables

- intentions to pursue a science-related research career (**5-point ordinal scale**)
- intentions to pursue a science-related research career (**binary**: “Probably yes” and “Definitely yes” =1)

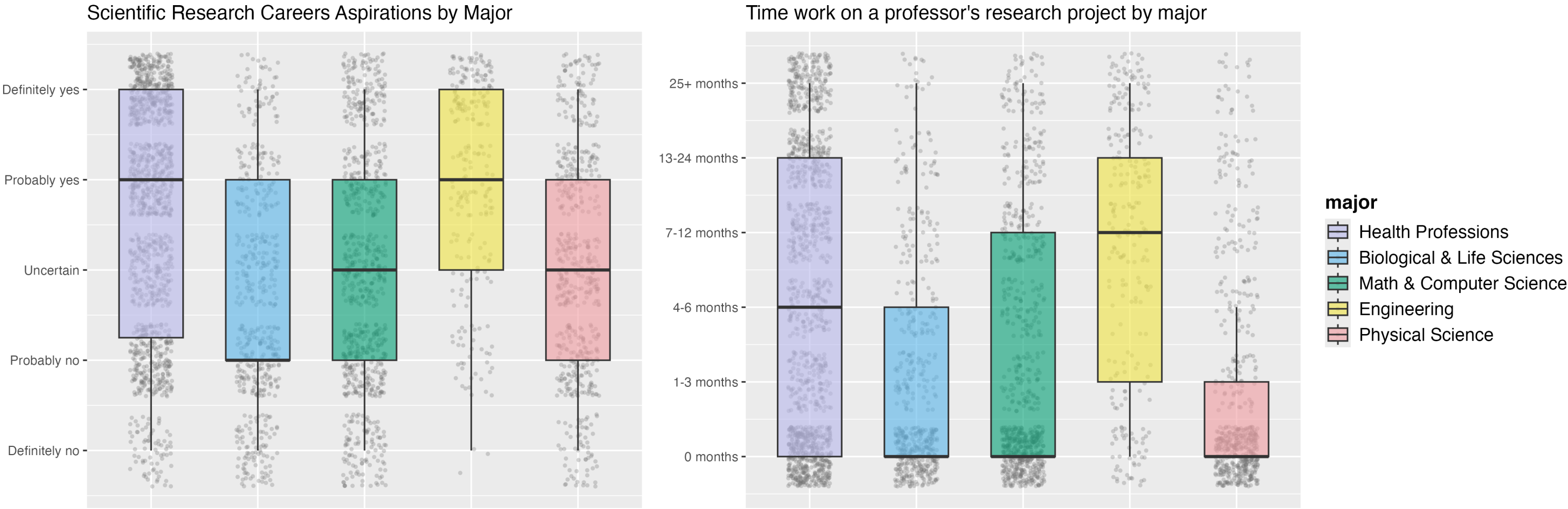
– Student-level predictors

- students **undergraduate research experience** (**6-point**, 1 = None to 6 = > 25 months)
- **psychological constructs**, i.e., science self-efficacy, science identity, academic self-concept, and habits of mind (**continuous**)
- **goal orientation variables**, i.e., aiming to make theoretical contributions to science, helping others who are in difficulty, and becoming an authority in the field (**4-point**)
- students' **major GPA** (**8-point**, 1 = D to 8 = A/A+)
- **college time allocation variable**, e.g., time spend on studying/homework, socializing with friends in person, using social media, etc, (**8-point**, 1 = None to 8 = >20 hours)

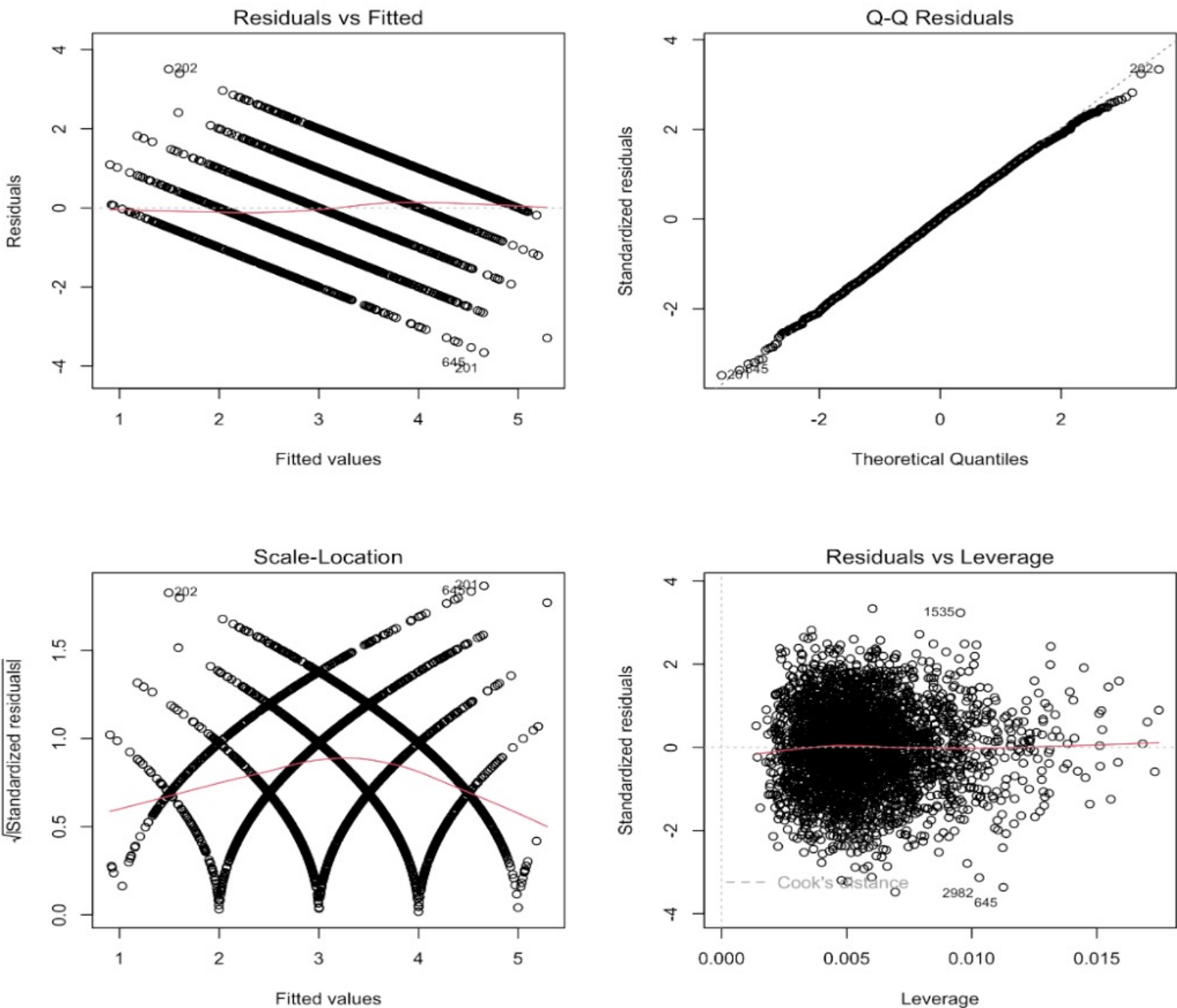
Methods

- **RQ1** Descriptive analysis; box plots
- **RQ2** OLS linear regression; kernel regression (Nadaraya-Watson)
- **RQ3** ML classification; predictor importance analysis

Findings RQ1 Box Plots



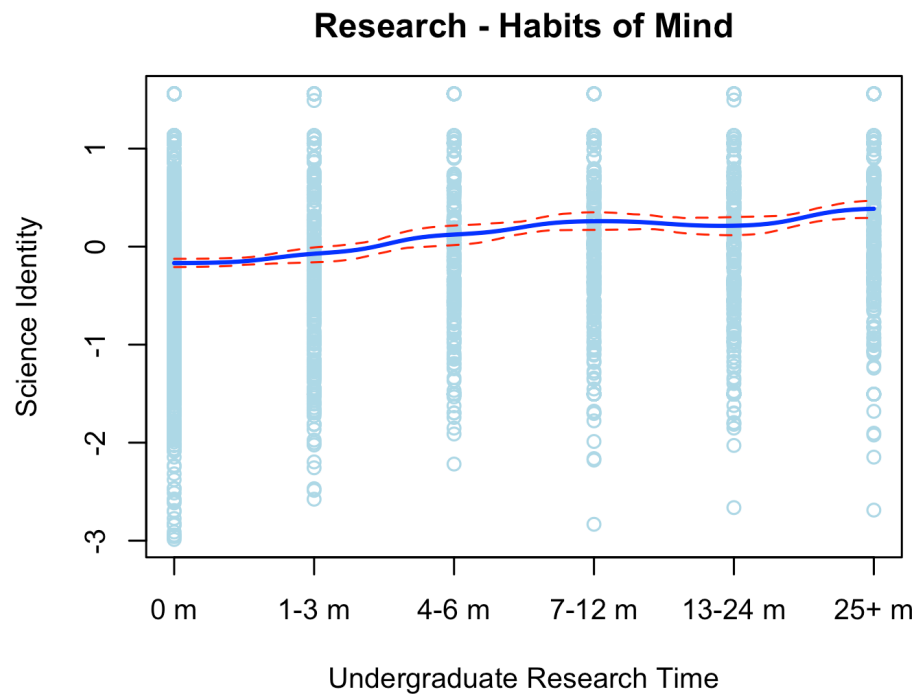
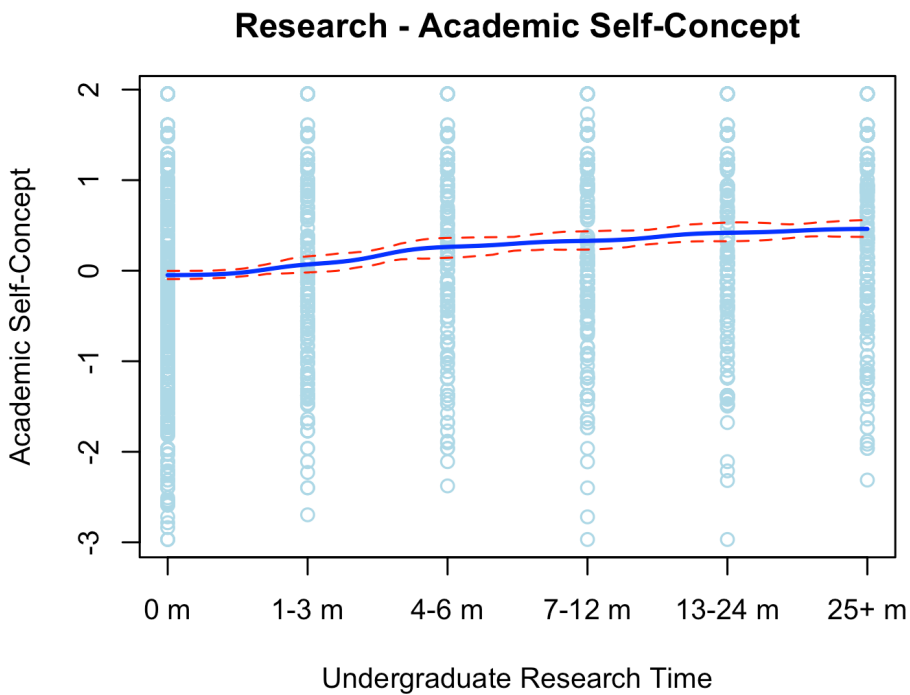
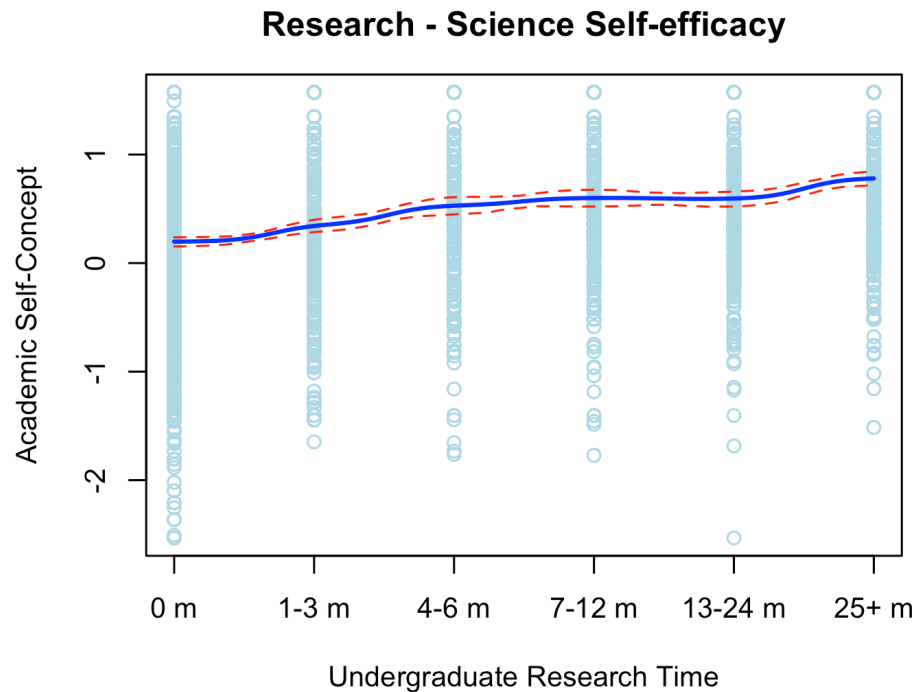
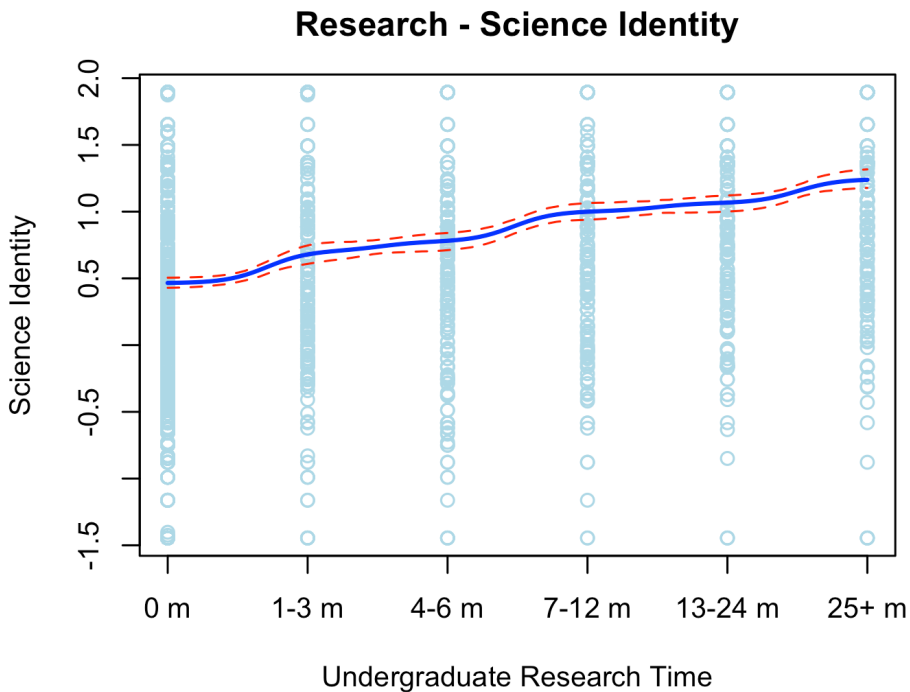
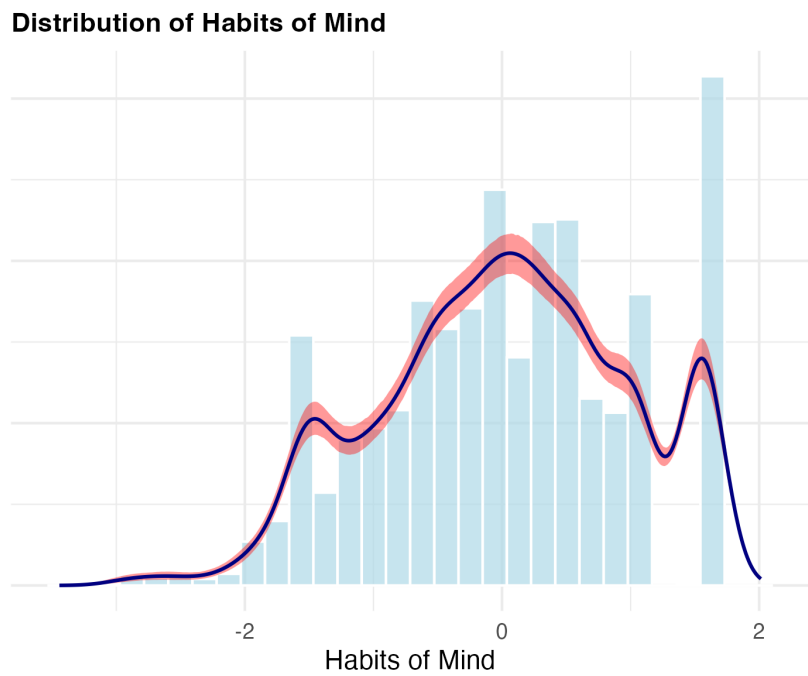
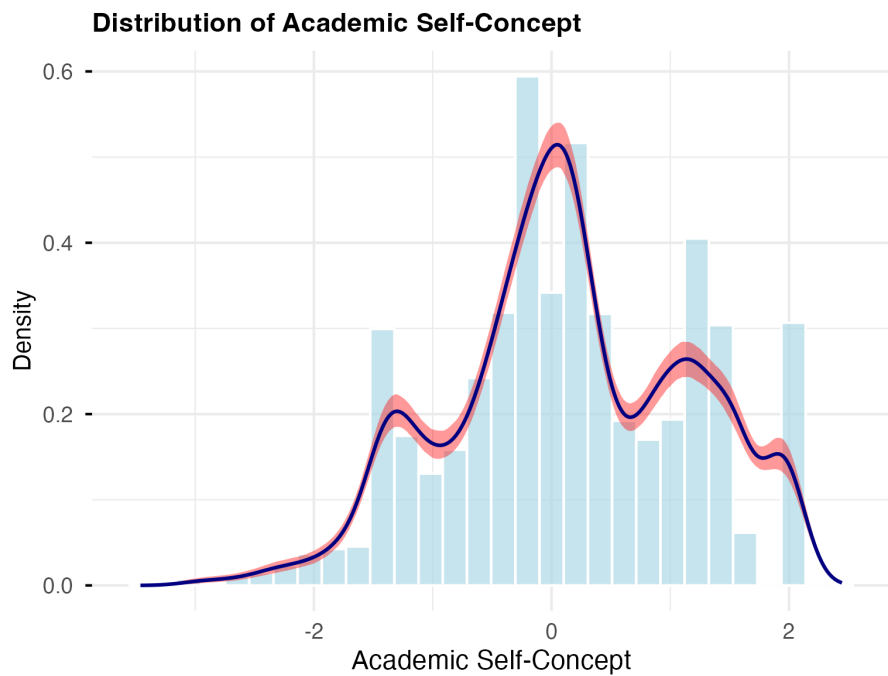
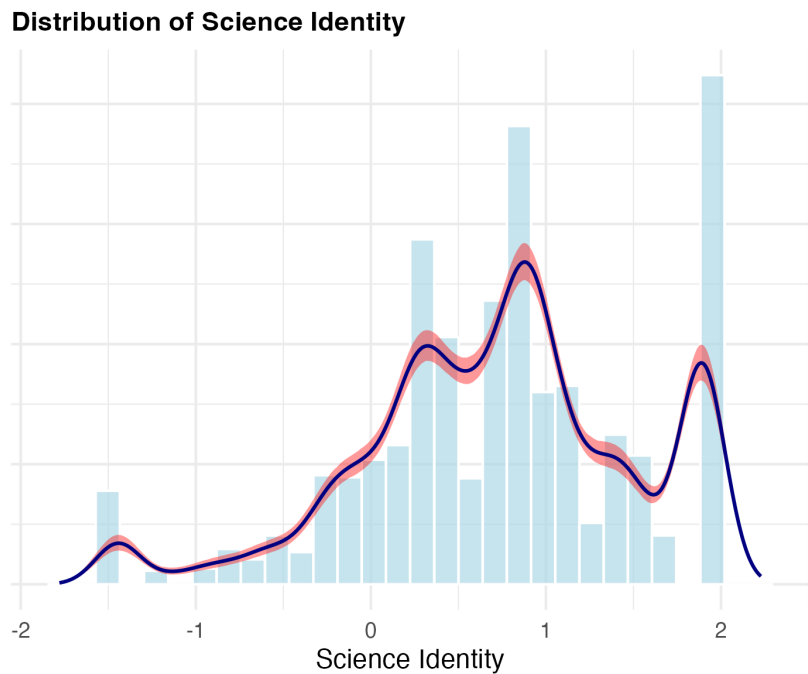
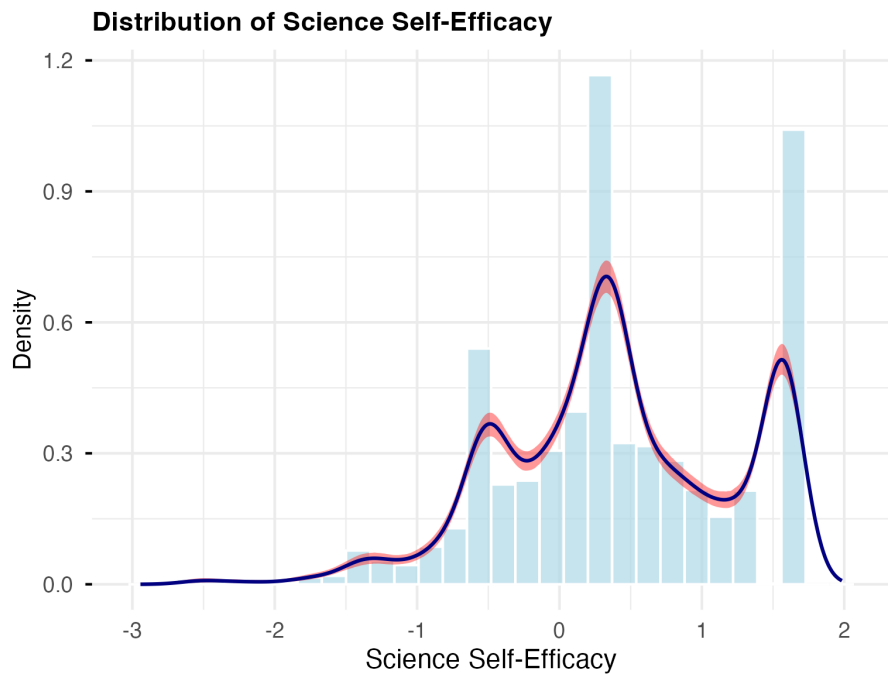
Findings RQ2 Linear Regression



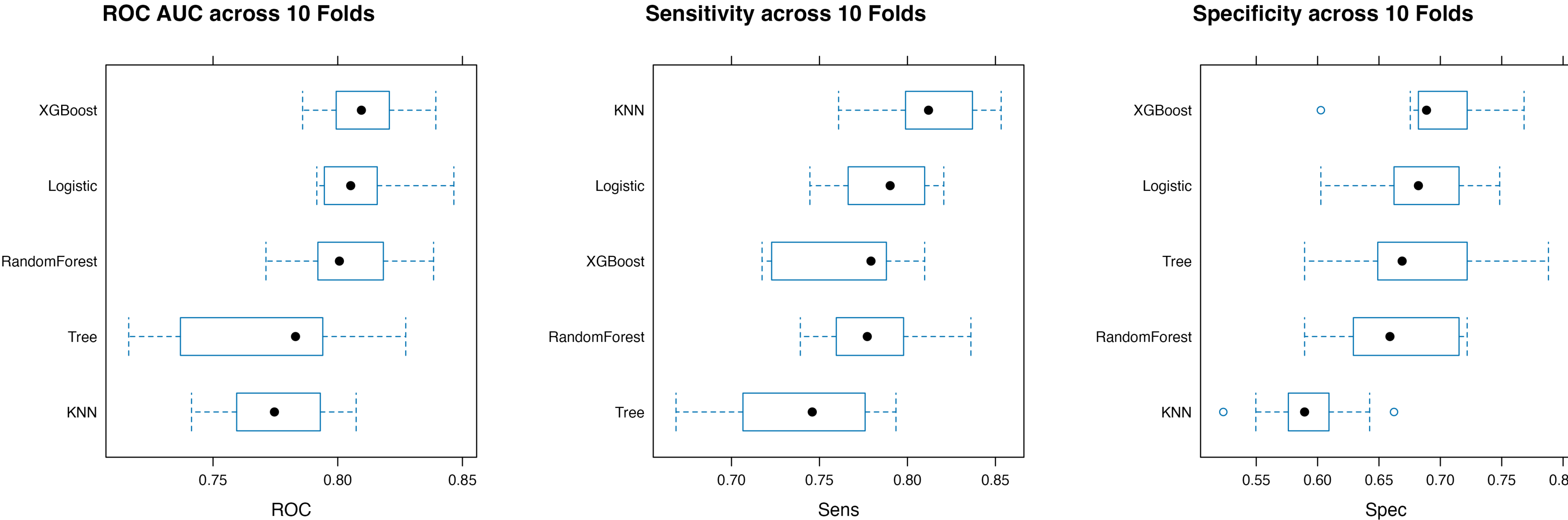
Variable	B	β	SE	95% CI [LCI, UCI]		p
(Intercept)	2.350	NA	0.166	2.020	2.680	***
Science identity (z)	0.615	0.366	0.032	0.554	0.677	***
Academic self-concept (z)	-0.111	-0.085	0.022	-0.154	-0.068	***
Make theoretical contribution to science (GOAL19)	0.370	0.277	0.023	0.324	0.416	***
Helping others who are in difficulty (GOAL11)	-0.117	-0.067	0.025	-0.167	-0.067	***
Undergraduate Research (UGR)	0.072	0.099	0.011	0.050	0.094	***
Studying/doing homework (HPW17)	0.037	0.041	0.014	0.009	0.065	**
Socializing with friends in person (HPW28)	-0.049	-0.057	0.013	-0.075	-0.022	***

Adjusted R² = 0.36

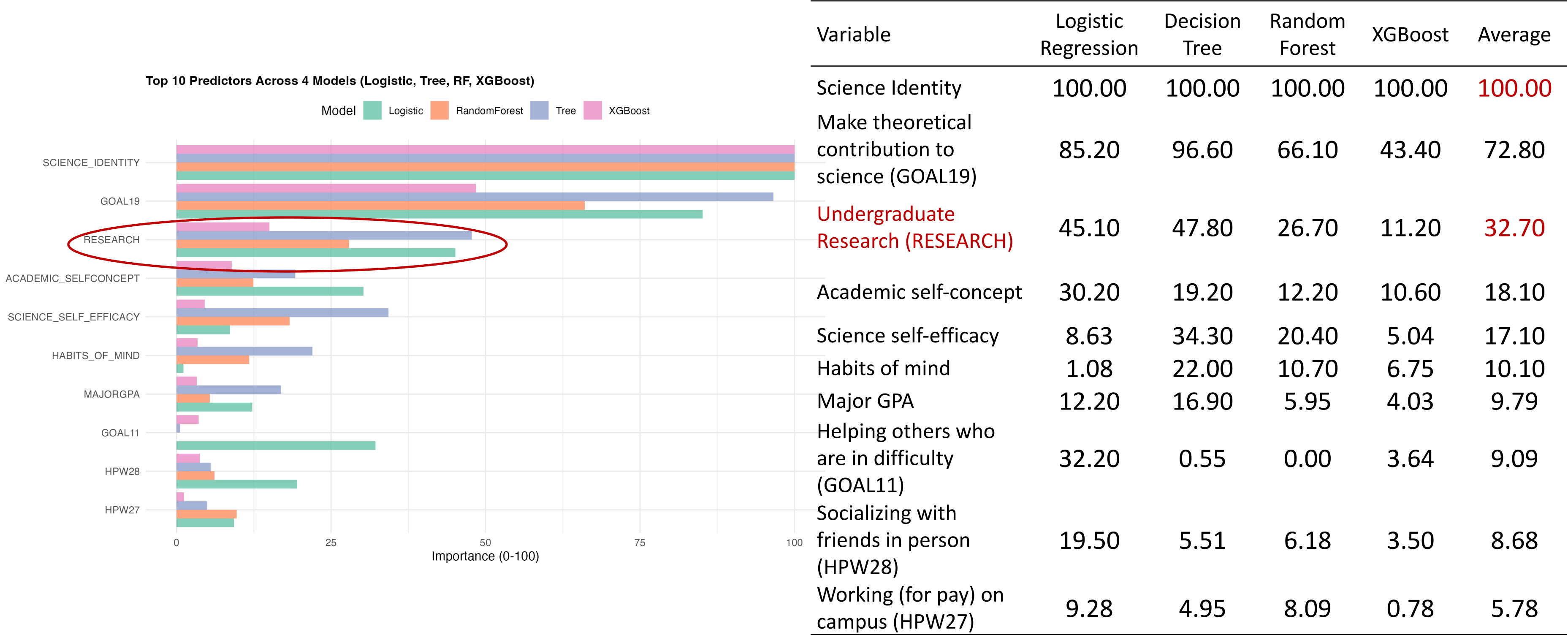
Findings RQ2 Kernel Density & Kernel Regression



Findings RQ3 ML Prediction



Findings RQ3 ML Variable Importance Rankings



Limitation and Future Directions

- **Limitation**
 - Cross-sectional nature of the CSS data
 - Self-reported data
 - Binary classification of career aspirations
- **Future Directions**
 - Quasi-experimental designs, e.g., PSM, IVs, sensitivity analysis
 - Longitudinal studies
 - Multilevel modeling to include institutional influence



Q & A

- Thanks for listening!
- Welcome questions and constructive feedback.

